

p -adic Variational Quantum Circuits Have No Barren Plateaus

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AIQxQIA 2026 | Perugia | 8 October 2026

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The obstruction we all work around

For a variational circuit $U(\theta)$ and observable M ,

$$\mathcal{L}(\theta) = \langle U(\theta)\psi_0, M U(\theta)\psi_0 \rangle.$$

McClean et al. (2018): if the ansatz forms a unitary 2-design,

$$\mathbb{E}[\partial_\theta \mathcal{L}] = 0, \quad \text{Var}[\partial_\theta \mathcal{L}] = O(2^{-n}).$$

The landscape is flat almost everywhere; the shot count to resolve a descent direction grows exponentially in the qubit count.

Mitigations attack the *circuit*: initialisation, local cost functions, shallow or structured ansätze. Every one of them keeps the scalars $\mathbb{C}$.

What does the theorem actually use?

Three ingredients, and none of them is quantum mechanics:

1. a **compact** ensemble, hence a Haar *probability* measure;
2. moment data from unitary ***t*-designs**;
3. “small” in the **Archimedean** absolute value.

All three are facts about $\mathbb{C}$.

So: what happens if we change the field?

Changing the scalars

Following Aniello, Mancini and Parisi, rebuild the kinematics over a quadratic extension K of the p -adic field $\mathbb{Q}_p$:

- states are vectors in K^N , $N = 2^n$;
- gates are unitaries for a K -valued Hermitian form;
- the geometry is **ultrametric**: $|x + y|_p \leq \max\{|x|_p, |y|_p\}$.

Why anyone builds this: ultrametric geometry is the intrinsic geometry of **hierarchical data** — taxonomies, phylogenies, file systems.

The pieces exist already: a p -adic qubit, a tensor product, a universal gate set, a first quantum neural network.

Three facts, and that is all the p -adics you need

1. Small means divisible

$|x|_p = p^{-v_p(x)}$, where v_p counts factors of p . At $p = 3$: $|18|_p = 1/9$, because $18 = 2 \cdot 3^2$.

2. The scale is discrete

Values live in $p^{\mathbb{Z}}$. A gradient is a unit, or divisible by p , or by p^2 — nothing in between. No continuum of “slightly small”.

3. Sums do not grow

A sum is as large as its largest term, *unless* leading digits cancel — and that is a condition modulo p : finite and checkable.

Asking the question correctly

$\text{Var} = O(2^{-n})$ has no p -adic meaning. The faithful translation asks how often the gradient is *as large as it can be*.

Question

Let G_N be the group of $N \times N$ unitaries with entries in the ring of integers $\mathcal{O}$, with Haar probability measure ν . For integral $M = M^*$, $A = -A^*$, and

$$U(\theta) = U_B \exp_p(p\theta A) U_A, \quad \theta \in \mathbb{Z}_p,$$

we always have $|\partial_\theta \mathcal{L}|_p \leq p^{-1}$.

With U_A drawn from ν : does $\Pr[|\partial_\theta \mathcal{L}|_p = p^{-1}]$ stay bounded away from 0 as $n \rightarrow \infty$, and for which pairs (M, A) ?

Surprise 1: the usual ensemble does not exist

Proposition

$U(K^N)$ is *non-compact* for every $N \geq 2$.

At $p = 3$ there is an explicit unbounded family of unitaries. The reason is structural: over a local field the form $\sum \bar{z}_i z_i$ is isotropic, and compactness in the complex case came from *positivity*, which the p -adic model gives up.

No Haar probability measure $\Rightarrow$ “Haar-random circuit” is meaningless $\Rightarrow$ the plateau question, posed over the full group, has no answer.

Fix: the integral unitary group $G_N = U_N(\mathcal{O})$ — compact, and it contains every circuit built from the known p -adic gate sets (Pauli, CNOT, $SO(3)_p$ gates, Hadamard when $\sqrt{2} \in K$).

Surprise 2: the benchmark is p^{-1} , not 1

The series $\exp_p(X)$ converges only for $\|X\| < p^{-1/(p-1)}$ — and because the value group is discrete, that is *equivalent* to $\|X\| \leq p^{-1}$.

So every convergent ansatz has $\exp_p(\theta X) \equiv 1 \pmod{p}$: one-parameter subgroups sit inside the principal congruence subgroup, and

$$|\partial_\theta \mathcal{L}|_p \leq p^{-1} \quad \text{always.}$$

“The gradient is a p -adic unit” asks for something **no convergent circuit can deliver**. The factor p in $\exp_p(p\theta A)$ is not a normalisation we chose — it is the whole content of the convergence condition.

The missing t -design combinatorics are not needed

Compactness of G_N comes with a filtration:

$$G_N \longrightarrow U_N(\mathcal{O}/p^k) \longrightarrow \cdots \longrightarrow U_N(\mathbb{F}_{p^2}).$$

- Haar averages over G_N push forward to **uniform counting** on finite unitary groups.
- Haar-random states push forward to the uniform measure on a finite Hermitian sphere.
- At level one the gradient's leading digit is $\langle \tilde{\psi}, \Gamma \tilde{\psi} \rangle \in \mathbb{F}_p$, a **character sum** we evaluate in closed form.

Counting in finite groups replaces the moment combinatorics the complex theory gets from t -designs.

Theorem

Let Γ be the commutator $[U_B^* M U_B, A]$ reduced modulo p . If Γ splits over $\mathbb{F}_p$ with no eigenvalue of multiplicity above $N - r$, then

$$\Pr\left[|\partial_\theta \mathcal{L}|_p = p^{-1}\right] \geq 1 - \frac{1}{p} - 10 p^{1-r}.$$

The bound does **not depend on n** . And $r = \Theta(2^n)$ generically, so the correction is *doubly* exponentially small exactly where the complex plateau is worst.

	Complex	p -adic
Ensemble	$U(2^n)$, compact	full group non-compact; use G_N
Moments	unitary t -designs	character sums over $U_N(\mathbb{F}_{p^2})$
Gradient	$\text{Var} = O(2^{-n})$	$\Pr[\text{maximal}] \geq 1 - p^{-1} - 10p^{1-r}$
Scaling in n	exponential decay	none
Bad set	full measure as $n \rightarrow \infty$	locus $[\hat{M}, A] \equiv 0 \pmod{p}$
Expressibility	more expressive, flatter	more random, larger gradients

The last row is the one to remember: over K , randomness is what *produces* large gradients.

A witness you would actually write down

Take $p \equiv 3 \pmod{4}$, so $K = \mathbb{Q}_p(i)$. On n qubits:

$$M = X_1, \quad A = -\frac{i}{2}(1 - Z_1).$$

In words: *measure Pauli X on the first qubit, rotate the first qubit about Z .*

Then $C = [M, A] = Y_1$, the hypothesis holds, and $r = 2^{n-1}$.

At $n = 10$, $p = 3$ the correction term is smaller than 3^{-500} :

$$\Pr[\text{gradient maximal}] \geq 1 - \frac{1}{3} - 3^{-500}.$$

What replaces the plateau

The obstruction does not vanish — it changes shape.

- **Complex:** a *measure* phenomenon. The bad set has full measure as $n \rightarrow \infty$; no choice of M and A escapes it.
- **p -adic:** a *locus*. The bad set is exactly $[\hat{M}, A] \equiv 0 \pmod{p}$ — check it by reducing two integer matrices and multiplying them.

It is small but **not null**: a clopen union of congruence classes of measure $O(p^{-1})$. Positive measure, a distinction the complex theory never has to make.

Design criterion: choose $\hat{M}$ and A that visibly fail to commute modulo p .

What this does *not* say

- **The splitting hypothesis is real.** Over a finite field a Hermitian matrix may have eigenvalues outside the fixed field — we give a 2×2 counterexample over $\mathbb{F}_9$. Outside that class: open.
- **Level one only.** We control $\Pr[v_p(\nabla \mathcal{L}) \geq 1]$, not the full valuation distribution.
- **Standing assumptions.** p odd, $K/\mathbb{Q}_p$ unramified, one parameter in a two-block ansatz.
- **No hardware.** No device computes over K . And there is no Monte Carlo readout: p -adic probabilities are not in $[0, 1]$, so readout is algebraic, digit by digit.

So why care about a model with no hardware?

1. **It is a diagnostic about the complex theory.** The plateau survives every change we make to the *circuit*, and does not survive the change we make to the *field*. That locates the hypothesis a mitigation has to attack: concentration in the Archimedean absolute value.
2. **It hands over a checkable criterion.** Non-commutation modulo p is a finite test, not an asymptotic statement.
3. **The machinery is reusable.** Haar $\rightarrow$ counting in finite unitary groups should serve other second-moment questions.

Barren plateaus are a theorem about the
Archimedean absolute value.

Change the field and they go.

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Backup: non-compactness at $p = 3$

For $N = 2$ the form $\bar{z}_1 z_1 + \bar{z}_2 z_2$ is isotropic over K : there is a non-zero vector of norm zero. Unitary groups of isotropic Hermitian forms over a local field are non-compact.

Concretely at $p = 3$ one exhibits unitaries whose entries have arbitrarily large absolute value — the group is unbounded, so no Haar *probability* measure exists on it. Restricting to integral entries is what restores compactness.

Backup: the counting lemma

For Γ selfadjoint over $\mathbb{F}_{p^2}$ and $c \in \mathbb{F}_p$, let

$$\mathcal{N}_c = \#\{u : \langle u, u \rangle = 1, \langle u, \Gamma u \rangle = c\}.$$

If Γ is r -non-degenerate, then

$$\left| \frac{\mathcal{N}_c}{|\mathbb{S}_N|} - \frac{1}{p} \right| \leq 10 p^{1-r}, \quad |\widetilde{\mathbb{S}}_N| = p^{2N-1} - (-1)^N p^{N-1}.$$

The value $c = 0$ is the barren-plateau case: the leading digit of the gradient vanishes.

Backup: why the splitting hypothesis is not free

Over $\mathbb{F}_9$, with $\iota^2 = -1$:

$$\Gamma = \begin{pmatrix} 0 & 1 + \iota \\ 1 - \iota & 0 \end{pmatrix}, \quad \chi_\Gamma(x) = x^2 + 1.$$

The roots are $\pm \iota \in \mathbb{F}_9 \setminus \mathbb{F}_3$: Hermitian, but not diagonalisable over the fixed field.

Closing this gap needs the classification of *pencils* of Hermitian forms over a finite field — the main open follow-up.

Backup: how large can r be?

Γ has $N = 2^n$ eigenvalues in a p -element field, so pigeonhole gives $m_{\max} \geq \lceil N/p \rceil$ and

$$r \leq N \left(1 - \frac{1}{p}\right).$$

A simple spectrum is impossible once $N > p$. Spread evenly, $r \approx N(1 - 1/p)$; the witness of the talk achieves $r = N/2$.

Either way $r = \Theta(2^n)$, so $10p^{1-r}$ is of order p^{-2^n} .